

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING)** Department of Computer Science and
Engineering

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	10% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	Narravula Navya Sri	Reg. No.:	192324220
Section / Batch:	AI & DS	Date:	26-8-26

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze the given questions thoroughly before answering.
- Compare multiple aspects such as methods, results, advantages, and limitations clearly.
- Critically evaluate the strengths, weaknesses, and implications with proper justification.
- Present meaningful insights, interpretations, and well-supported conclusions from the comparisons.

Question Type	Focus Area	Questions
Comparative Analysis	Comparative analysis of Dynamic Programming and Greedy algorithms for optimization problems.	Q1

SECTION A – Comparative Analysis

Given word lengths and line width, solve Word Wrap using DP and compare with Job Sequencing problem solved using Greedy differences in objective functions (minimizing raggedness vs maximizing profit). Evaluate correctness of both approaches. Com computational complexity. Discuss real-world applications. Generate insights on optimization strategies.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Comparative Analysis					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%				

Comparison & Differentiation	25%				
Critical Evaluation	25%				
Synthesis & Insight Generation	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



SIMATS Online Programming Lab
DAA PROGRAMMING

NABHRAVULA NAVYA SRI
PRO31AC208A

ASSESSMENT CO4-AT3-COMPARATIVE ANALYSIS

Language: Python C C++ Java

QUESTIONS

Q1
Word Wrap Problem vs Job Sequencing with Deadlines (DP vs Greedy)
100 marks

1/7 answered

Q1 Word Wrap Problem vs Job Sequencing with Deadlines (DP vs Greedy) 100 marks

Given word lengths and line width, solve Word Wrap using DP and compare with Job Sequencing problem solved using Greedy. Analyze differences in objective functions (minimizing raggedness vs maximizing profit). Evaluate correctness of both approaches. Compare computational complexity. Discuss real-world applications. Generate insights on optimization strategies.

Your Code

```
def word_wrap_dp(words, max_width):
    n = len(words)
    dp = [float('inf')] * (n + 1)
    next_split = [0] * n
    dp[0] = 0
    for i in range(n - 1, -1, -1):
        current_width = -1
        for j in range(i, 0):
            current_width += len(words[j]) + 1
            if current_width <= max_width + 1:
                spaces = max_width - (current_width - 1)
                cost = spaces**2 if j < n - 1 else 0
                if cost + dp[j + 1] < dp[i + 1]:
                    dp[i] = cost + dp[j + 1]
                    next_split[i] = j + 1
            else:
                break
        lines = []
        curr = 0
        while curr < n:
            end = next_split[curr]
            lines.append(" ".join(words[curr:end]))
            curr = end
    return dp[0], lines

class Job:
    def __init__(self, id, deadline, profit):
        self.id = id
        self.deadline = deadline
        self.profit = profit

def job_sequencing_greedy(job):
```

Input

stdin input

Output

Min Raggedness Cost: 2
Lines:
This is an
example of
word wrap